

Yong Zane

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WORK EXPERIENCES

Advanced Remanufacturing and Technology Centre (ARTC), A*STAR

Research Engineer Full Time

Oct 2024 – Present

- Architected a modular pipeline for 2D engineering drawings utilizing VLMs; de-coupled textual/numerical inference streams to eliminate class-aliasing errors and boosted automated ingestion volume by 40%.
- Designed a microservices API Gateway in ASP.NET for secure multi-tenant auth and routing; ensured data isolation and stable operations across 10+ internal/external manufacturing teams.
- Engineered FastAPI workers using RabbitMQ and Pydantic for strict schema validation; improved system resilience by de-coupling VLM inference from the request cycle and ensuring data contract integrity.
- Implemented a Docker and Kubernetes-backed CI/CD pipeline for model deployment, achieving a 12× reduction in deployment time by enforcing environment consistency across stages.
- Developed a responsive Next.js/React dashboard for real-time VLM telemetry; enabled stakeholders to visualize and verify complex metadata with zero-latency UI updates.

AWARDS

NTU CAO x ST Engineering Hackathon 2024

Apr 2024

First Runner-up

- Developed a data-driven surgery risk assessment system using traditional machine learning in scikit-learn, enabling more informed tool usage decisions and reducing surgical downtime and costs.
- Applied domain knowledge to make an intentional design decision in a machine learning risk prediction pipeline, prioritizing clinical interpretability and usability.

NTU EEE MLDA Deep Learning Week 2024 Hackathon (Group Leader)

Mar 2024

First Runner-up

- Implemented a zero-shot video quality evaluation pipeline using small pre-trained transformers, enabling stable convergence and reducing compute energy usage by 50% during fine-tuning of a large text-to-video model.
- Led a 4-person team, ensuring effective task ownership and execution, resulting in a second-place award.

RESEARCH PUBLICATIONS

[A Multi-Stage Hybrid Framework for Automated Interpretation of Multi-View Engineering Drawings Using Vision Language Model](#) [International Conference on Industrial Engineering and Applications (ICIEA)], 2025

[From drawings to decisions: A hybrid vision-language framework for parsing 2D engineering drawings into structured manufacturing knowledge](#) [Robotics and Computer-Integrated Manufacturing], 2025

[Automated Parsing of Engineering Drawings for Structured Information Extraction Using a Fine-tuned Document Understanding Transformer](#) [IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)], 2025

EDUCATION

Nanyang Technological University, Singapore

Master of Science (Artificial Intelligence)

Aug 2026 – Aug 2028 (Expected)

- On-going (Part-time)

Bachelor of Engineering (Bioengineering)

Jul 2020 – May 2024

- Graduate with Honours (Distinction)
- MOE Tuition Grant Holder

SKILLS

Frontend

- HTML, CSS, JavaScript, TypeScript
- React, Angular, Next.js

Backend

- Python, C++
- Node.js, .NET

Databases

- MySQL, PostgreSQL

Cloud Platforms

- AWS, Firebase

AI/ML frameworks and libraries

- PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, HuggingFace